

CUSTOMER CHURN ANALYSIS FOR TELECOM INDUSTRY

1. Introduction

In the highly competitive telecom sector, the cost of acquiring a new customer significantly outweighs the cost of retaining an existing one. This project was initiated to analyze telecom transactional data, build a predictive framework for customer churn, and derive actionable, data-backed retention strategies. The goal was to transition from reactive customer service to predictive, AI-driven lifecycle management.

2. Abstract

A comprehensive analytics pipeline was engineered utilizing Python and MySQL. A synthetic dataset of 10,000 telecom records was generated and piped through a relational database to sanitize missing records and engineer aggregate usage features (such as the *Frustration Index*). A Random Forest binary classification model was deployed to predict churn probability with high ROC-AUC accuracy. Game Theory algorithms (SHAP) were applied to demystify the model, proving that short-term contracts and network instability were the primary catalysts for churn. Finally, customers were segmented into 'At Risk', 'Loyal', and 'Dormant' cohorts to optimize marketing spend.

3. Tools Used

- **Python (Pandas, Numpy, Scikit-learn, Matplotlib/Seaborn):** Deployed for data generation, machine learning (Random Forest), probability scoring, and generating presentation visuals.
- **SHAP (Shapley Additive exPlanations):** Utilized for advanced ML explainability to isolate the specific variables driving individual customer churn.
- **MySQL Workbench:** Utilized as the data processing engine to manage NULL values and engineer complex behavioral metrics (Call Volume, Frustration Index) via aggregate views.

4. Steps Involved in Building the Project

- **Step 1: Data Generation:** Engineered an enterprise-scale dataset in Python, embedding real-world telecom logic (e.g., higher network drops directly increase churn probability) and introducing missing values to test downstream data engineering.
- **Step 2: Database Ingestion & Feature Engineering:** Piped raw data into MySQL. Executed SQL UPDATE queries to impute missing contract types. Created dynamic relational views to calculate average monthly call times and a weighted Frustration Index.
- **Step 3: Machine Learning & SHAP Analysis:** Extracted the engineered SQL data back into Python. Trained a Random Forest classifier, evaluating performance via an ROC-AUC curve. Applied SHAP tree explainers to identify feature importance globally.
- **Step 4: Segmentation & Deliverables:** Programmed conditional logic to segment the user base based on AI churn probabilities and SQL behaviors. Exported visualizations and constructed a strategic PowerPoint presentation and strategic recommendation brief.

5. Conclusion

The integration of SQL data engineering and Python machine learning provides a definitive blueprint for telecom retention. The SHAP analysis mathematically proves that churn is primarily driven by unresolved network friction and the lack of long-term contract incentives. By leveraging the AI's output to segment the customer base, telecom operations can now deploy highly targeted interventions—subsidizing 5G upgrades for 'At Risk' users and deploying win-back campaigns for 'Dormant' users—effectively reducing overall churn and protecting top-line revenue.